1

Exercise 1.3. A telephone number consists of ten digits, of which the first digit is one of $1, 2, \ldots, 9$ and the others can be $0, 1, 2, \ldots, 9$. What is the probability that 0 appears at most once in a telephone number, if all the digits are chosen completely at random?

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- 10 digit phone number.
- 1^{st} digit $\in \{1, \ldots, 9\}$ → can't be zero.
- Remaining digits $\in \{0, \ldots, 9\}$.

⇒ Find $P(0 \text{ appears at most once})$

Step 1: Total number of valid phone numbers

- The 1^{st} digit must be 1-9 → 9 options
- Each of the remaining 9 digits can be 0-9 → 10 options per digit

Total possible phone numbers $= 9^1 \cdot 10^9$

Step 2: Count favorable Outcomes

0 appears at most once (ie. 0 times or exactly 1 time)

- Case 1: zero appears 0 times
  - 1^{st} digit: 9 options (1-9)
  - Remaining 9 digits: only digits 1-9 (no zero) → 9 options per digit

Number of phone numbers with no zeros $= 9^1 \cdot 9^9 = 9^1 \cdot \binom{9}{0} \cdot 9^9$

- Case 2: Zero Appears Exactly Once
  - 1^{st} digit: 9 options (1-9)
  - Remaining 9 digits: zero must appear in one of the 9 digits. (2-10)

    $\binom{9}{1} = 9$

    Other 8 digits: 9^8

Number with exactly one zero $= 9 \cdot 9 \cdot 9^8 = 9^{10} = 9^1 \cdot \binom{9}{1} \cdot 9^8$

Step 3: Total Favorable Cases

Total Favorable $= (9^1 \cdot 9^9) + (9 \cdot 9 \cdot 9^8) = 2 \cdot 9^{10}$

Step 4: Probability

$$\boxed{P = \frac{\text{Favorable}}{\text{Total}}} = \frac{2 \cdot 9^{\cancel{10}\,9}}{\cancel{9} \cdot 10^9} = \frac{2 \cdot 9^9}{10^9} = 2 \cdot \left(\frac{9}{10}\right)^9$$

$$P = \frac{\text{Favorable}}{\text{Total}} = 0.7748$$

(2)

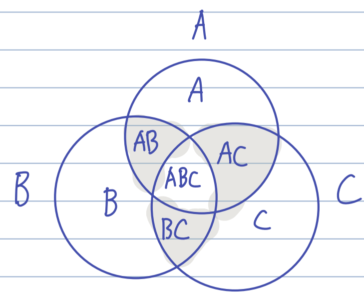
Exercise 1.6 (Skills Exercise). Events A, B, and C are defined in a sample space Ω. Find expressions for the following probabilities in terms of $P(A), P(B), P(C), P(AB), P(AC), P(BC)$, and $P(ABC)$; here AB means $A \cap B$, etc.:

(a) the probability that exactly two of A, B, C occur;

- AB occur, but not C : $P(ABC') \rightarrow \underline{A\cap B\cap C'}$
- AC occur, but not B : $P(ACB') \rightarrow \underline{A\cap B'\cap C}$
- BC occur, but not A : $P(BCA') \rightarrow \underline{A'\cap B\cap C}$

$P(\text{exactly 2 occur}) = [P(AB) - P(ABC)] + [P(AC) - P(ABC)] + [P(BC) - P(ABC)]$

$P(\text{exactly 2 occur}) = P(AB) + P(AC) + P(BC) - 3P(ABC)$

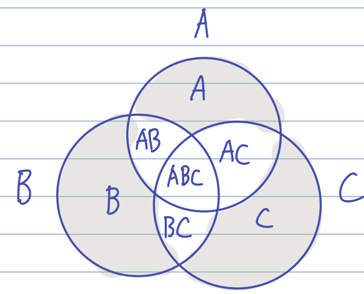


(b) the probability that exactly one of these events occurs;

- A only occurs : $P(AB'C') \rightarrow \underline{A\cap B'\cap C'}$
- B only occurs : $P(BA'C') \rightarrow \underline{A'\cap B\cap C}$
- C only occurs : $P(CA'B') \rightarrow \underline{A'\cap B'\cap C}$

$P(A \text{ only}) = P(A) - P(AB) - P(AC) + P(ABC)$
$P(B \text{ only}) = P(B) - P(AB) - P(BC) + P(ABC)$
$P(C \text{ only}) = P(C) - P(AC) - P(BC) + P(ABC)$

$P(\text{exactly 1 occurs}) = P(A) + P(B) + P(C) - 2P(AB) - 2P(AC) - 2P(BC) + 3P(ABC)$



(c) the probability that none of these events occur.

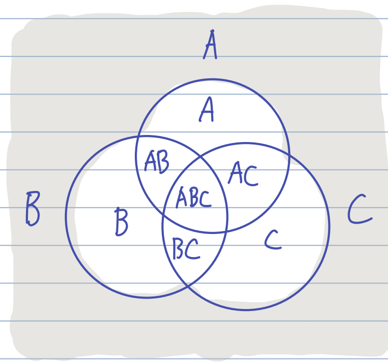
- $P(A'\cap B'\cap C') = 1 - P(\text{At least one of } A, B \text{ or } C)$
  $P(A\cup B\cup C)' = 1 - P(A\cup B\cup C)$

$P(\text{None of ABC occur}) = 1 - [P(A \text{ only}) + P(B \text{ only}) + P(C \text{ only})] - [P(AB) + P(BC) + P(AC) - 2P(ABC)]$

$= 1 - [P(A) + P(B) + P(C) - 2P(AB) - 2P(AC) - 2P(BC) + 3P(ABC)] - [P(AB) + P(BC) + P(AC) - 2P(ABC)]$

$= 1 - P(A) - P(B) - P(C) + \cancel{2}P(AB) + \cancel{2}P(AC) + \cancel{2}P(BC) - \cancel{3}P(ABC) - \cancel{P(AB)} - \cancel{P(BC)} \cancel{-P(AC)} \cancel{+2P(ABC)}$

$P(\text{None of ABC occur}) = 1 - P(A) - P(B) - P(C) + P(AB) + P(AC) + P(BC) - P(ABC)$



√ $\underline{A'\cap B'\cap C'}$: None of ABC occur

X $A'\cup B'\cup C'$: At least one of ABC did not occur

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Exercise 1.9. In which of the following are events A and B mutually exclusive? → (a)

(a) Roll two dice. A is the event of a sum of 9; B is the event of a double (i.e., the same value on both dice).

A: (3,6) (6,3) (4,5) (5,4)

B: (1,1) (2,2) ⋮ (6,6)

"Can Both Happen?" NO.

$A \cap B = \emptyset$

No overlapping values, hence A and B are mutually exclusive. ✓

(b) Draw 13 cards from a deck of 52 cards. A is the event of drawing at least one club; B is the event of drawing no aces.

"Can Both Happen?" Yes.

A: At least one club

B: No aces

Overlapping possibility from drawing Ex): 1 club, 0 aces,
hence A and B are not mutually exclusive.

(c) Toss a coin twice. A is the event of a head on the first toss; B is the event of a head on the second toss.

A: H

B: H

} (H,H)

"Can Both Happen?" Yes.

Overlapping possibility from flipping both heads,
hence A and B are not mutually exclusive.

④

Exercise 1.13. The letters in the word FULL are rearranged at random. What is the probability that it still spells FULL?

$$\boxed{\text{Permutation Formula with Repeated} = \frac{n!}{k_1! \cdot k_2!}}$$

- $n!$: total # of letters
- $k!$: counts of each repeated letter.

Step 1: Total number of letters arrangement

$$\text{Total Permutations} = \frac{4!}{\underbrace{2!}_{\text{Two repeated L's}}} = 12$$

- Why not 2? (FUL_1L_2, FUL_2L_1)
  - Same arrangements in real life, therefore count as one single unique permutations.

Step 2: Count favorable Outcomes = 1

Amog the 12 letters arrangements, only one is "FULL".

Step 3: Total Favorable Cases = 1

Step 4: Probability

$$\boxed{P = \frac{\text{Favorable}}{\text{Total}}} = \frac{1}{12}$$

$$P = \frac{\text{Favorable}}{\text{Total}} = \frac{1}{12}$$

5

Exercise 1.16. Twenty cookies are to be distributed to ten children. In how many ways can the cookies be distributed if (a) any cookie can be given to any child; (b) If two cookies are to be given to each child?

- Assume cookies and children are distinct.

a) $\underset{r=1}{\overset{\text{1st cookie}}{10}} \cdot 10 \cdot 10 \cdot 10 \ldots \underset{r=20}{10}$

Possible each child gets more than one cookie $= 10^{20}$
(Not limited by number of cookies)

b) - All Possible orderings: $20!$
  - All unique combinations, order overcounted.
  - Double counting pairs. Ex): A/B, B/A

- Adjust for indistinguishable groups of 2 cookies/child: $(2!)^{10}$

Each child: $2!$
\# of groups: 10
$\underset{r=1}{2!} \cdot 2! \cdot 2! \ldots \underset{r=10}{2!}$

- Adjust for indistinguishable cookie order overcounted: $10!$
There are $10!$ ways to reorder these groups, should not count.

\# of ways to form 10 unordered pairs:

$$= \frac{20!}{(2!)^{10} \cdot 10!}$$

internal ordering of each pair
reordering of 10 pairs

(A J)
(B I)
(C H)
(D G)
(E F)
(F E)
(G D)
(H C)
(I B)
(J A)
⋮
~~(J A)~~
~~(I B)~~
~~(H C)~~
⋮

Two Cookies for Every Child $= \frac{20!}{(2!)^{10} \cdot \cancel{10!}} \cdot \cancel{10!}$ ways to assign 10 pairs to 10 children

Two Cookies for Every Child $= \frac{20!}{(2!)^{10}}$

With Replacement
Number of ways to choose r objects from n distinct objects if they same subject could be chosen repeatedly:
n^r

Without Replacement
Number of ways to arrage n distinct object with order of arrangement matters:
$n!$